

Senuka Dinuwara

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🌐 senuka-dinuwara 🔄 Senuka-Dinuwara

Profile

Motivated Computer Science undergraduate with hands-on experience in Python, Java, and database management. Skilled in architecting local applications, optimizing data flows, and solving complex problems through practical backend development. Eager to contribute to software engineering roles with a strong learning mindset and a commitment to high-quality, scalable code.

Work Experience

Password Security Tools

- Developed PassGuard, a Java 22 and Maven-based desktop password manager utilizing PBKDF2 with HMAC-SHA256 for secure user authentication and passkey verification. Architected a local-first vault that secures credential storage using AES-GCM encryption, featuring a real-time entropy-based password strength checker and an automated password generator. Designed a modular Java Swing UI separating backend security services from presentation layers, complete with safe clipboard integration and encrypted file import/export utilities.

Timetable Maker

- Developed Timetable Maker, a Java Swing desktop app that automates academic scheduling using a greedy conflict-free algorithm to resolve overlaps across faculty availability, room capacities, and student cohorts. Engineered a modular architecture featuring bulk CSV data import, priority-based constraint handling, and a multi-view dashboard with HTML schedule export.

DataPilot (Lightweight File-Based Data Framework)

- Architecting DataPilot, an open-source, lightweight Python data management framework designed to provide a unified API for CRUD operations across file-based storage engines.
- Developed a modular core routing engine that dynamically matches data payloads to CSV and JSON connectors based on file extensions.
- Currently building a foundational validation layer, maintaining an automated test suite using PyTest, and packaging the initial framework for distribution via PyPI (pip install).

Coursework Project | SDG Awareness Site (NextGen Industries)

- Created Splash Screen and Volunteer pages using HTML, CSS, and JavaScript as part of a group project.

Education

BSc (Hons) Computer Science, University of Westminster

2024 – present

- Second Year (Level 5)
- Expected Graduation: 2028

London

Foundation Program, Informatics Institute of Technology

- Achieved a Distinction pass

2023 – 2024

Colombo, Sri Lanka

Ordinary Level (O/L) Examination, Sumedha College

2011 – 2023

- Completed Ordinary Level (O/L) exams with top grades in ICT (A), English (A), and Mathematics (B).
- Active member of the ICT Club and Photography Club.

Gampaha, Sri Lanka

Skills

Programming Languages

- Python, Java, HTML, CSS

Frameworks & Tools

- Tkinter, SQL, Git

Problem-Solving & Debugging

Project Management & Team Collaboration

Certificates

- Critical Thinking and Problem Solving
- Python Practice: Object-Oriented Programming
- Build real world applications with Python

References

W.W.Resta Priyanthi Fernando, *Visiting lecturer - University of Kelaniya*
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Dewmini Ruwanpathirana, *Software Engineer - IFS R&D International (pvt) Ltd*
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